



Soil carbon and nitrogen stocks in nitrogen-fertilized grass and legume-grass forage systems

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Abstract Soils under grasslands contribute significantly to terrestrial carbon (C) and nitrogen (N) pools, and grassland management practices directly impact pool size through effects on plant biomass. Our objective was to quantify above and belowground plant biomass and soil C and N accumulation for year-round forage systems based on N-fertilized grasses (GN; 290 kg N ha⁻¹ yr⁻¹) or legume-grass mixtures (LG; 30 kg N ha⁻¹ yr⁻¹) when grazed or harvested for hay. Soil C and N accumulation (0–0.4 m) were

evaluated before and 6 yr after imposing treatments. Average root-rhizome mass did not differ for GN and LG systems (3690 and 3020 kg OM ha⁻¹, respectively). Aboveground residual biomass was 55% greater for GN than LG (4890 vs. 3150 kg OM ha⁻¹, respectively) and 46% greater for grazed than hayed swards (4770 vs. 3270 kg OM ha⁻¹, respectively) at the end of the warm season. Aboveground C and N pools were 70 and 58% greater for GN than LG, respectively. After 6 yr, soil C and N concentration (0–0.1 m) increased 6.7 and 0.8 g C kg⁻¹ and 0.6 and 0.2 g N kg⁻¹, for GN and LG, respectively. Annual C accumulation (0–0.1 m) was 1.3 and 0.2 Mg C ha⁻¹

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for GN and LG systems, respectively. Thus, a year-round forage system based on N-fertilized grasses accumulated more plant biomass and surface soil C and N than a legume-grass mixture, with the grass system advantaged by presence of a perennial C4 species and application of moderate to high N fertilizer rates.

Keywords Carbon accumulation · C3 legumes · C4 grasses · Grassland management practices · Nitrogen accumulation · Nitrogen fertilization

Introduction

Grasslands occupy an estimated 3.6 billion ha globally and account for approximately one-fourth of C stored in the world's soils (Follett and Reed 2010), acting as important sinks of atmospheric carbon dioxide (CO₂) released from global deforestation and land-use changes (Conant et al. 2017). Agricultural practices that result in soil disturbance (i.e., tillage) impact soil organic C (SOC) accumulation and stability (Cambardella and Elliott 1992; Six et al. 1999), whereas improved grassland management practices that increase above- and belowground biomass production often result in increases in SOC (Conant et al. 2001, 2017; Allard et al. 2007; Ammann et al. 2007). These practices include use of legumes and perennial forages, N fertilization, and improved grazing management in livestock systems (Conant et al. 2017). Additionally, integrating legumes into grass-based systems can increase SOC (Tarre et al. 2001; Cotrufo et al. 2013) through improved forage production and addition of N to the system through biological fixation.

Conant et al. (2017) observed that SOC sequestration rates were positive in most studies in which fertilizer was applied, averaging 0.82 and 0.54 Mg C ha⁻¹ yr⁻¹ for organic and inorganic fertilizer use, respectively, in a review of 126 papers. Providing 200 kg N ha⁻¹ yr⁻¹ as inorganic N or liquid cattle manure on a temperate grassland combined with four harvests by clipping per season increased net C sequestration by 1.47 Mg ha⁻¹ yr⁻¹, whereas no fertilization and less frequent cutting (three times per year) resulted in a net C loss of 0.57 Mg ha⁻¹ yr⁻¹ (Ammann et al. 2007). Moderate

to heavy grazing resulted in loss of soil C (average of 6.7%) while light grazing contributed to an 0.8% increase in soil C in a meta-analysis of 115 studies (Zhou et al. 2017). Harvest method may also affect soil C, as SOC (0–0.6 m) in Georgia, USA increased almost three times more when bermudagrass [*Cynodon dactylon* (L.) Pers.] was grazed than when mechanically harvested (Franzluebbers et al. 2001). Other studies in the eastern USA reported greater SOC accumulation in grazed than hayed swards (Franzluebbers and Stuedemann 2009; Sochorová et al. 2016), and responses were often attributed in part to greater nutrient return in grazed swards contributing to increased soil fertility (Franzluebbers et al. 2000).

Greater biodiversity and plant functional diversity increase resistance to disturbance and can stabilize grassland productivity (Isbell et al. 2015; Tracy et al. 2018), but the relationship of species richness and soil C accumulation is inconsistent in the literature. Addition of highly productive species, those with specific rooting characteristics (i.e., high belowground biomass), or legumes is thought to have the greatest positive effect on soil C accumulation (De Deyn et al. 2009, 2011; Skinner and Dell 2016). In Texas, USA, SOC concentration (0–0.15 m) increased only 58% as much under N-fertilized bermudagrass–annual ryegrass (*Lolium multiflorum* Lam.; 350 kg N ha⁻¹ yr⁻¹) pasture as for a bermudagrass–clover (*Trifolium* spp.) mixture receiving no N fertilizer over 19 yr (Wright et al. 2004). In a semiarid region of China, forage legume monocultures established in abandoned cropland increased soil C accumulation by 0.64 Mg C ha⁻¹ yr⁻¹ more than grass monocultures (Liu et al. 2017). Santos et al. (2019) reported that integration of rhizoma perennial peanut (*Arachis glabrata* Benth.) increased by 22% soil particulate organic matter concentrations for the 0.15-m soil depth, although no differences were observed for soil C or N stocks among forage treatments (bahiagrass and rhizoma peanut monocultures and bahiagrass–rhizoma peanut mixtures). Jensen et al. (2012) concluded that although studies with legumes are limited, SOC concentrations can be increased by legume inclusion in pastures, especially in permanent pastures and with perennial legume species.

Although previous studies evaluated the impacts of grassland management on soil C and N responses (Allen et al. 2006; Jensen et al. 2012; Cong et al. 2014), relatively few studies compared the benefits of

year-round N fertilized grasses vs. grass-legume forage systems on C and N allocation in above- and below-ground pools and associated soil C and N responses. Based on existing literature, we hypothesized that integrating forage legumes with grasses in year-round forage systems would increase soil C and N stocks to a similar or greater degree than N-fertilized grass systems that received moderate to high levels of N fertilizer, and that utilization by grazing would result in greater soil C and N than by haying. The objective of this study was to quantify above and belowground plant biomass and soil C and N accumulation for year-round forage systems based on N-fertilized grasses or legume-grass mixtures when grazed or harvested for hay during 6 yr.

Materials and methods

Site description, treatments and experimental design

The experiment was conducted at the Plant Science Research and Education Unit (PSREU) at Citra, FL (29.24°N, 82.10°W). Predominant soils are Inceptisols (sandy, siliceous, hyperthermic Typic Humaquepts) and Entisols (hyperthermic, uncoated Typic Quartzipsamments). During the 5 yr preceding the start of the experiment, the area was conventionally tilled each year and planted to annual row crops including sweet corn (*Zea mays* L.), peanut (*Arachis hypogaea* L.), and cotton (*Gossypium hirsutum* L.). Initial soil pH (0–0.15 m) was 6.3, and Mehlich-3 P, K, and Mg concentrations were 128, 41, and 46 mg kg⁻¹, respectively. The 30-yr total annual rainfall at the site averages 1300 mm. During study years, annual precipitation averaged 1270 mm. There were large deviations from the average in 2016 and 2018 when 923 and 1570 mm of rain were recorded, respectively.

Treatments were two forage systems (N-fertilized grass [GN; 290 kg N ha⁻¹ yr⁻¹]; and legume-grass mixtures [LG; 30 kg N ha⁻¹ yr⁻¹]) and two defoliation methods (grazing and mechanically harvested for hay). Treatments were arranged in three replications of a randomized complete block with a split-plot arrangement. Forage systems were allocated to mainplots and defoliation methods to subplots. Treatments were imposed from 2011 through 2018. Each

experimental unit (subplot) was 92 m², and forage systems were composed of a perennial warm-season perennial forage component overseeded yearly with a mixture of annual cool-season species (Table 1). In 2011, ‘Florigrass’ rhizoma peanut rhizomes (1200 kg ha⁻¹) and ‘Tifton 85’ bermudagrass [T85; *Cynodon dactylon* (L.) Pers. × *Cynodon transvaalensis* Burt Davy] aboveground stems (1460 kg ha⁻¹) were planted in the LG and GN mainplots, respectively. From 2011 through 2017, cool-season species were overseeded each year in mid-October or early November in both LG and GN mainplots. Prior to overseeding, GN plots were clipped to 0.10 m stubble height to remove excess herbage mass to allow effective seed-soil contact for cool-season forage seeds. A no-till drill was used to plant ‘Florida 401’ rye (*Secale cereale* L.; 64 kg pure live seed (PLS) ha⁻¹) and ‘Prine’ annual ryegrass (*Lolium multiflorum* Lam; 20 kg PLS ha⁻¹) in plots of both systems, while LG plots were also planted to ‘Dixie’ crimson (*Trifolium incarnatum* L.; 11 kg PLS ha⁻¹) and ‘Southern Belle’ red (*T. pratense* L.; 7 kg PLS ha⁻¹) clovers.

Defoliation and fertilization management

Defoliation intensity and frequency were matched to each treatment combination (forage system × defoliation management) and season. During June through October, both grazing and hay harvest events occurred when bermudagrass canopies were 0.45- to 0.50-m tall and rhizoma peanut canopies were 0.20- to 0.25-m tall. Target stubble heights were 0.15 to 0.20 m for T85 bermudagrass and 0.10 m for rhizoma peanut under grazing (Ortega et al. 1992; Hernández-Garay et al. 2004; Liu et al. 2011), and 0.10 m under hay management for both systems. From January through April, cool-season forages in both grazed and hayed treatments were defoliated to a 0.10-m stubble every 5 to 6 wk, depending on herbage accumulation rates and the reproductive stage of the grasses. Generally, there were three defoliation events during the winter and a range of three to six events during the summer, with more harvest events for grazed than hay treatments and for GN than LG treatments. The last defoliation event of the seasons occurred in mid-April for winter and in early October for summer. For grazed plots, mob stocking was used. The number of animals assigned per plot varied from 6 to 10, depending on

Table 1 Description of year-round forage systems used

Forage system	Perennial warm- season component	Annual cool-season components	N fertilizer rate
Legume (LG)	‘Florigraze’ rhizoma perennial peanut	Annual ryegrass, rye and clovers	30 kg ha ⁻¹ yr ⁻¹
N-fertilized grass (GN)	‘Tifton 85’ bermudagrass	Annual ryegrass and rye	290 kg ha ⁻¹ yr ⁻¹

pre-grazing herbage mass, and grazing time ranged from 30 to 120 min until target stubble height was achieved. Animals used were dry cows, cow-calf pairs, and yearlings (*Bos spp.*), depending upon availability. Hay plots were clipped using a flail mower, and all harvested forage was removed from the plots at each defoliation event.

At the beginning of the warm season in May, GN and LG plots were fertilized with 50 kg K ha⁻¹ based on soil test. The LG plots received no N fertilizer during the summer, while all GN plots received 50 kg N ha⁻¹ in May and an additional 50 kg N ha⁻¹ after each of the first three defoliation events of summer for a total of 200 kg N ha⁻¹. To aid establishment, both GN and LG plots received 30 kg N, 13 kg P, and 50 kg K ha⁻¹ 4 wk after planting the cool-season species. No additional fertilizer was applied to LG plots during the winter, but GN plots received 30 kg N ha⁻¹ after each of the first two defoliation events for a total of 90 kg N ha⁻¹. Total N fertilizer inputs were 30 and 290 kg N ha⁻¹ yr⁻¹ for LG and GN systems, respectively.

Weeds in LG plots during the summer were controlled using Impose (ammonium salt of imazapic), 2,4-D (2,4-dichlorophenoxyacetic acid), and clethodim [(±)2-[(E)-1-[3-chloroallyloxyimino]propyl]-5-[2-(ethylthio)propyl]-hydroxycyclohexen-2-one (1)] at 300, 600, and 1200 mL ha⁻¹, respectively. Chaparral (potassium salt of 2-pyridine carboxylic acid, 4-amino-3,6-dichloro-; metsulfuron methyl) (200 mL ha⁻¹) was used to control broadleaf weeds in GN plots. During the experimental period, herbicide applications occurred two to three times annually, and the herbicide products were those commonly used by producers and were determined based on predominant weed species in the forage system. Supplemental irrigation was applied on a limited basis for cool-season forage establishment or growth, typically once or twice per cool season (10–13 mm application⁻¹) during severe spring or fall drought.

Plant sampling and analyses

Treatments were imposed for 6 yr, and soil responses were characterized prior to treatment imposition and after 6 yr. Plant pool mass and nutrient composition, however, were assessed only when the systems had been in place for several years, i.e., in the last 2 or 3 yr of treatment imposition. Herbage accumulation and botanical composition data for these experimental units was published previously (Silva et al. 2021). In the current paper, they are mentioned briefly to provide context for results reported herein. Readers are referred to Silva et al. (2021) for detailed consideration of these responses and associated sampling methodology.

Residual herbage biomass and C and N concentrations

Aboveground residual biomass (RB) was defined as the sum of stubble and litter (as defined by Allen et al. 2011) at the time of sampling. Sampling occurred twice per year after defoliation events (grazing or cutting) at the end of the cool season (5 May 2016 and 2017) and the end of the warm season (3 Nov. 2016 and 15 Nov. 2017). Two 0.25-m² sites were sampled to soil level per plot at each date, and the sample was dried in a forced-air oven at 60 °C for 72 h, screened to remove soil contamination, redried overnight, and weighed. A composite of the two subsamples per plot was ground in a Wiley mill (Model 4, Thomas-Wiley Laboratory Mill, Thomas Scientific) to pass a 1-mm stainless steel screen. Organic matter (OM) concentration was determined as described by Moore and Mott (1974) and C and N concentrations by dry combustion in a Flash EA 1112 N/C analyzer (Thermo Scientific, Delft, Netherlands) after samples were processed in a mixer mill (MM 400, Retsch GmbH, Germany). Carbon and N pools were calculated by multiplying RB by C and N concentrations, and pools are expressed in kg (C or N) OM ha⁻¹.

Root-rhizome biomass and C and N concentrations

Root-rhizome biomass (RRM) was quantified on 3 Nov. 2015, 8 Dec. 2016, and 15 Dec. 2017. Three (0.10-m diameter by 0.20-m deep) cores were taken randomly per plot at each sampling date. Cores were composited within an experimental unit, and roots and rhizomes were washed to remove soil over a 2-mm mesh screen. Samples were dried at 60 °C until constant weight and weighed. The composite sample was ground using a Wiley mill (Model 4, Thomas-Wiley Laboratory Mill, Thomas Scientific) to pass a 1-mm stainless steel screen, and a subsample was subsequently ashed at 500 °C for 4 h. Root-rhizome biomass is expressed on an OM basis. A separate subsample was analyzed for C and N concentrations as described for residual herbage biomass samples. Concentrations were multiplied by RRM to calculate C and N pools.

Soil sampling and analyses

Soil sampling occurred on 7 March 2011 (pre-treatment) and 9 Feb. 2017 (after imposing treatments for 6 yr). In each experimental unit, four undisturbed cores were randomly sampled to 0.40-m depth using plastic liners (50 mm in diameter). After sampling, liners and their contents were immediately frozen and were later cut into 0- to 0.10-, 0.10- to 0.20-, and 0.20- to 0.40-m depth increments. Soil was removed from liners, sieved through a 2-mm screen and oven-dried at 60 °C for 7 d. Subsamples were ball milled for 5 min using a MM 400 ball mixer mill (Retsch, Haan, Germany) prior to determining total C and N using a Carlo Erba NA1500 elemental analyzer (Thermo Scientific, Delft, Netherlands). Soil bulk density was determined using the undisturbed core method (Grossman and Reinsch 2002) and drying samples at 105 °C. Soil C and N stocks (Mg ha^{-1}) were calculated considering C and N concentration, soil bulk density and depth (Ussiri et al. 2006). Data are reported in concentrations (g C or N kg^{-1} soil) and stocks (Mg C or N ha^{-1}) for prior to (baseline), and after 6 years, of treatment imposition. The magnitude of change observed across the 6 yr was calculated as 2017 concentration or stock of C or N minus the initial (2011) values.

Statistical analysis

Data were analyzed using R software (R Core Team 2018). For plant responses, a linear mixed model was used with the function “lme” where forage system, defoliation method, and their interaction were considered fixed effects, and block and year were random. Least-squares means were obtained using the “emmeans” package (Lenth et al. 2019) and reported for significant main and interaction effects for plant responses ($P \leq 0.05$) and an unstructured correlation structure was used. Soil responses were analyzed within depths using a linear mixed model with forage system, defoliation management, and their interactions as fixed effects and block as random. All effects and interactions for soil responses were considered significant at $P \leq 0.1$ because these responses are often highly variable, making changes due to management difficult to assess (Dubeux et al. 2006a; Skinner et al. 2006). Treatment means were separated using Tukey’s test.

Results

Aboveground residual biomass and C and N pools

For samples taken at the end of the warm season, aboveground RB and C and N pools were affected by forage system and defoliation method (Table 2). The GN system had 55% greater RB than LG (4890 vs. 3150 kg OM ha^{-1} , respectively), 71% greater C pool in the RB (1910 vs. 1120 kg ha^{-1} , respectively), and 58% greater N pool (87 vs. 55 kg ha^{-1} , respectively) (Table 2). Grazed swards had 46% greater RB than hayed swards (4770 vs. 3270 kg OM ha^{-1} , respectively), 51% greater C pool in RB (1830 vs. 1210 kg C ha^{-1}), and 63% greater N pool (88 vs. 54 kg N ha^{-1} , respectively) (Table 2).

At end of the cool season, aboveground RB was affected by forage system ($P = 0.012$) and the defoliation management effect approached significance ($P = 0.090$), but there was no effect of either system or defoliation management on C and N pool sizes (Table 2). Aboveground RB was 51% greater for GN than LG systems (4980 vs. 3300 kg OM ha^{-1} , respectively), and means were 4700 versus 3580 kg OM ha^{-1} , respectively, for grazing and hay management ($P = 0.090$). The C and N pools in RB at the end

Table 2 Aboveground residual biomass, C:N ratio, and C and N pools for forage systems at the end of the warm and cool seasons

Forage system [†]	Warm season				Cool season			
	Biomass kg ha ⁻¹	N pool	C pool	C:N	Biomass kg ha ⁻¹	N pool	C pool	C:N
GN	4890	87	1910	23	4980	89	1750	20
LG	3150	55	1120	20	3300	67	1360	20
SE	547	12.8	237	1.1	421	9.6	139	0.38
<i>P</i> value [‡]	0.002	0.010	0.002	0.252	0.012	0.156	0.457	0.195
<i>Defoliation method</i>								
Grazing	4770	88	1830	21	4700	85	1600	20
Hay	3270	54	1210	22	3580	71	1520	21
SE	531	11.6	235	1.1	395	9.0	131	1.0
<i>P</i> value [‡]	0.001	< 0.001	0.004	0.246	0.090	0.375	0.634	0.133

[†]GN = N-fertilized grass system; LG = legume-grass mixture system
[‡]*P* value for comparison between forage systems or between defoliation methods

of the cool season did not differ due to either system or defoliation method and averaged 1560 kg C ha⁻¹ and 78 kg N ha⁻¹ (Table 2). In both seasons, C/N ratio of aboveground RB remained under 23, which favors nutrient mineralization (Table 2).

Root-rhizome mass and C and N pools

Root-rhizome mass was not affected by defoliation management ($P = 0.644$) or forage system ($P = 0.123$). Average root-rhizome mass for GN and LG plots was 3690 and 3020 kg OM ha⁻¹, respectively. Similar root-rhizome mass led to similar belowground C ($P = 0.586$) and N pool ($P = 0.436$) responses between forage systems, averaging 1180 kg C ha⁻¹ and 49 kg N ha⁻¹. All treatments had root-rhizome C/N ratios of ~ 25.

Soil C and N concentrations and stocks six years after imposing treatments

Initial C and N concentrations (average of 7.7 g C kg⁻¹ and 0.4 g N kg⁻¹; 0- to 0.40-m depth) were relatively constant with depth (Table 3). Initial C and N stocks (0–0.40 m) averaged 43.8 Mg C ha⁻¹ and 2.5 Mg N ha⁻¹. After 6 yr of treatment imposition, soil C (0–0.10 m) concentration ($P = 0.075$) and stock ($P = 0.095$) were affected by forage system (Table 4). The GN system had 64 and 53% greater soil C concentration and stock, respectively, than LG (14.3 vs. 8.3 g C kg⁻¹ and 17.8 vs. 11.6 Mg C ha⁻¹, respectively; Table 4). Similarly, soil N concentration ($P = 0.096$) and stock ($P = 0.104$) were 62 and 49% greater for GN than LG systems (Table 5). No

differences in treatment responses were observed in deeper (> 0.10 m) soil depths (Table 5). Although initial soil C and N concentrations were similar across depths (Table 3), at the end of the 6-yr study, losses of C and N were observed at greater depths ($P < 0.001$) (Tables 4 and 5). After 6 yr, there were no effects of forage system ($P = 0.705$) or management strategy ($P = 0.407$) on soil bulk density when compared to initial values. Initial bulk density for 0- to 0.1-, 0.1 to 0.2, and 0.2 to 0.4 m were 1.3, 1.3 and 1.6 Mg m⁻³, while after 6 yr they averaged 1.4, 1.6 and 1.8 Mg m⁻³, respectively.

Changes in soil C and N concentrations and stocks during six years of imposing treatments

Changes in soil C concentration ($P = 0.080$) and stocks ($P = 0.091$) during the 6 yr in the 0- to 0.10-m depth were affected by forage system. Both GN and LG resulted in increases in C concentration (average of 6.3 and 0.8 g kg⁻¹ for GN and LG systems, respectively) (Table 4). Similarly, soil C stock (0–0.10 m) increased ($P = 0.091$) with time, with cumulative increases over 6 yr of 7.6 and 1.14 Mg C ha⁻¹ for GN and LG systems, respectively. Within the 0- to 0.10-m depth, rate of accumulation over 6 yr averaged 1.3 and 0.2 Mg C ha⁻¹ yr⁻¹, respectively, for GN and LG systems. Change in soil N concentration was also affected by forage system ($P = 0.097$) with a greater increase observed for GN than LG systems (0.6 and 0.2 g kg⁻¹, respectively; Table 5). There were no system differences for change in N stock, which averaged 0.7 and 0.3 Mg N ha⁻¹ for GN and LG systems, respectively.

Table 3 Initial soil C and N concentrations and stocks by depth

Treatment [†]	Soil depth (m)					
	0–0.10 C concentration (g kg ⁻¹)	0.10–0.20	0.20–0.40	0–0.10 C stock (Mg ha ⁻¹)	0.10–0.20	0.20–0.40
GN Graze	8.3	8.4	7.6	11.0	11.7	24.3
GN Hay	7.2	6.9	7.0	10.0	8.9	21.7
LG Graze	7.8	7.7	7.2	9.2	9.3	22.9
LG Hay	8.5	7.8	7.7	10.8	9.6	25.6
	N concentration (g kg ⁻¹)			N stock (Mg ha ⁻¹)		
	GN Graze	0.5	0.5	0.4	0.7	0.7
	GN Hay	0.4	0.4	0.4	0.6	0.5
	LG Graze	0.4	0.5	0.4	0.5	0.5
	LG Hay	0.5	0.5	0.4	0.7	0.6

[†]GN = N-fertilized grass system; LG = legume-grass mixture system; Graze – forage defoliated by grazing; Hay – forage defoliated by mechanical harvesting

Table 4 Soil C concentration and stock after 6 yr of treatment imposition and magnitude of change during the 6 yr as affected by soil depth and forage system

Soil depth (m)	C concentration (g kg ⁻¹)				C stock (Mg ha ⁻¹)			
	Forage system [†]				Forage system [†]			
	GN	LG	SE	P value [‡]	GN	LG	SE	P value [‡]
<i>After 6 yr of treatment imposition</i>								
0–0.10	14.3	8.7	1.3	0.075	17.8	11.6	1.72	0.095
0.10–0.20	10.8	10.8	2	0.978	14	13.6	2.74	0.931
0.20–0.40	7.3	6.5	1.45	0.740	23.1	21	4.5	0.77
<i>Magnitude of change during 6 yr</i>								
0–0.10	6.26	0.81	1.16	0.080	7.64	1.14	1.49	0.091
0.10–0.20	2.79	3.42	2	0.837	3.44	4.35	2.58	0.818
0.20–0.40	– 0.08	– 0.77	1.47	0.772	– 0.48	– 2.69	4.86	0.778

[†]GN = N-fertilized grass system; LG = legume-grass mixture system

[‡]P value for comparison between forage systems within a depth

Table 5 Soil N concentration and stock after 6 yr of treatment imposition and magnitude of change during the 6 yr as affected by soil depth and forage system

Soil depth (m)	N concentration (g kg ⁻¹)				N stock (Mg ha ⁻¹)			
	Forage system [†]				Forage system [†]			
	GN	LG	SE	P value [‡]	GN	LG	SE	P value [‡]
<i>After 6 yr of treatment imposition</i>								
0–0.10	1.1	0.7	0.09	0.096	1.3	0.9	0.13	0.104
0.10–0.20	0.8	0.7	0.14	0.938	1.0	0.9	0.19	0.895
0.20–0.40	0.5	0.4	0.09	0.644	1.5	1.3	0.29	0.626
<i>Magnitude of change during 6 yr</i>								
0–0.10	0.58	0.19	0.10	0.097	0.71	0.26	0.13	0.135
0.10–0.20	0.28	0.30	0.14	0.909	0.35	0.38	0.18	0.902
0.20–0.40	– 0.08	– 0.77	1.47	0.723	0.17	0.01	0.30	0.737

[†]GN = N-fertilized grass system; LG = legume-grass mixture system

[‡]P value for comparison between forage systems within a stratum

There were no treatment differences at depths > 0.10 m, and GN and LG soil C ($P = 0.818$) and N ($P = 0.902$) stocks in the 0.10- to 0.20-m depth averaged 0.57 versus 0.73 Mg C ha⁻¹ yr⁻¹ and 0.058 versus 0.063 Mg N ha⁻¹ yr⁻¹, respectively.

When considering total C stock change in the combined 0- to 0.20-m soil depth over the 6 yr, 31% was attributable to change in the 0.10 to 0.20-m depth for the GN system, but 79% was attributable to this depth for LG (Table 4).

In the 0.2 to 0.4-m depth, losses (negative changes in stock) occurred over the 6 yr for all responses, except change in N stock for GN (Tables 4 and 5). There was no effect of forage system or defoliation management on soil C stock in the 0.2- to 0.4-m depth ($P = 0.155$). Changes in C stock in the 0.2 to 0.4-m stratum over the 6 yr averaged $-3.77 \text{ Mg C ha}^{-1}$ for the haying treatment and $+0.61 \text{ Mg C ha}^{-1}$ under grazing.

Considering the cumulative 0.40-m soil depth sampled, initial soil C and N stocks averaged $43.8 \text{ Mg C ha}^{-1}$ and $2.47 \text{ Mg N ha}^{-1}$ prior to imposing treatments (Table 3). After 6 yr, there were no forage system ($P = 0.452$ for C; $P = 0.366$ for N) or defoliation method effects ($P = 0.819$ for C; $P = 0.846$ for N) for observed C and N stocks. Carbon and N stocks averaged $50.8 \text{ Mg C ha}^{-1}$ and 3.5 Mg N ha^{-1} , respectively, which represents an increase across the 6 yr of 7 Mg C ha^{-1} and 1 Mg N ha^{-1} . Magnitude of change in C stock for the 0- to 0.40-m soil depth over the 6-yr period was not affected by system ($P = 0.572$) or defoliation management ($P = 0.936$). Likewise, change in N stock for the 0- to 0.40-m depth was not affected by forage system ($P = 0.351$) or defoliation management ($P = 0.659$).

Discussion

Expansion of legume-based grassland–livestock systems has been described as an essential component of more sustainable and competitive ruminant production systems (Lüscher et al. 2014; Schultze-Kraft et al. 2018). Quantitative studies of legume introduction on SOC accumulation are limited (Jensen et al. 2012) especially in warm climates (Sollenberger et al. 2019), but the literature consensus is integration of forage legumes increases rate of soil C sequestration (Jensen et al. 2012; Cong et al. 2014). In this experiment, we evaluated sward characteristics and changes in soil C and N for year-round forage systems based on N-fertilized grass or legume-grass mixtures, when grazed or harvested for hay. We found the N-fertilized grass system, based on a perennial C4 bermudagrass and fertilized with $290 \text{ kg N ha}^{-1} \text{ yr}^{-1}$, accumulated soil C and N in the 0- to 0.10-m soil depth at a greater rate during 6 yr of treatment imposition than a legume-grass system based on the perennial legume,

rhizoma peanut, while defoliation method had no effect on soil N and C accumulation. This response was associated with greater biomass production for N-fertilized systems that may have favored return and more rapid incorporation of SOC into the surface soil layer. Thus, under the conditions of this study and with the species used, we rejected our hypotheses that (1) integrating forage legumes with grasses in year-round forage systems increases soil C and N stocks to a similar or greater degree than N-fertilized grass systems receiving moderate to high levels of N fertilizer, and (2) defoliation by grazing increases soil C and N to a greater degree than haying management.

Plant biomass inputs affecting soil C and N accumulation

Our first objective was to compare above and below-ground plant biomass for the four treatments. Annual herbage accumulation of these systems was described previously by Silva et al. (2021) and found to be greater for GN than LG (9890 vs. 5800 kg ha^{-1} ; $P = 0.01$, $SE = 1124$), but there was no effect of defoliation management. Aboveground RB favored the GN system, a response likely caused by greater annual herbage accumulation of GN than LG systems (Silva et al. 2021) and likely greater stubble mass as well. These responses can be attributed to presence of a perennial C4 grass receiving significant inputs of N fertilizer in GN ($290 \text{ kg N ha}^{-1} \text{ yr}^{-1}$) versus a C3-dominated system with much lower N inputs for LG ($30 \text{ kg N ha}^{-1} \text{ yr}^{-1}$). Litter decomposition rates were similar between systems (Silva et al. 2021) and did not contribute to the differences observed between them in aboveground RB. The greater C and N pools in RB for GN than LG systems were driven primarily by differences in amount of RB instead of composition of RB. Defoliation management also affected RB and its C and N pools, but only at the end of the warm season. Grazed swards had greater RB due to taller post-defoliation stubble height than hayed swards during the warm season and perhaps in part to greater amounts of plant damage associated with trampling and other grazing-related, plant-animal interactions.

Quantity of combined stubble and litter mass (referred to as aboveground residual biomass) has not been reported by others, but herbage accumulation and litter mass increased in response to increasing N fertilizer rates for C4 grasses including bahiagrass

(*Paspalum notatum* Flügge) (Dubeux et al. 2006b) and T85 bermudagrass (Liu et al. 2011). Greater litter mass has also been reported during the warm season for a bahiagrass monoculture, receiving either 0 or 50 kg N ha⁻¹ yr⁻¹, than bahiagrass-rhizoma peanut mixtures (Kohmann et al. 2018). In that experiment, greater litter mass of the grass was attributed to slower litter decomposition rates for bahiagrass than for the mixture. This was not the case in the treatments in the current experiment (Silva et al. 2021). Differences between forage systems for herbage accumulation and RB at the end of the cool season were likely associated with greater N fertilizer input to GN and carryover of recalcitrant C4 grass litter from summer GN swards. Thus, while previous studies accounted only for litter mass (senescent material on the soil surface) and not aboveground stubble mass, their results showed increasing litter mass associated with greater N fertilizer rates, taller stubble heights, and grass monocultures over grass-legume mixtures. These results support those of the current study.

Root-rhizome biomass did not differ among treatments. Belowground biomass is often considered an important driver of soil C accumulation (Sokel et al. 2019), but lack of difference in RRM and belowground C and N pools suggest it was not the principal driver of soil responses in the current study. Quantities of RRM were generally less than previously reported for T85 bermudagrass (Liu et al. 2015), similar to those reported for Florigrass rhizoma peanut pastures (Mullenix et al. 2016), and less than those reported for other rhizoma peanut cultivars (Dubeux et al. 2017; Shepard et al. 2018). Variation in reported root-rhizome mass of rhizoma peanut is large, however, reflecting differences due to cultivar, stand age, and management practices. For both above- and belowground plant material, C/N ratio was ~ 25. Quality of the plant material entering the system plays an important role in decomposition and is often defined by physical and chemical composition characteristics, especially C, N, and lignin concentrations (Thomas & Asakawa 1993). Generally, warm-season grasses have low N concentrations and high C/N ratios (> 40) (Vendramini et al. 2007) which may lead to net N immobilization (Thomas & Asakawa 1993). In this context, rate of N fertilizer has an important role not only on the quantity but also the quality of plant material incorporated into GN systems. However, these inputs are accompanied by economic costs and

greenhouse gas emissions associated with production, distribution and use of inorganic N fertilizer (Lal 2004).

Soil C and N responses to contrasting management strategies

Changes in soil C and N concentration and stock in the top 0.10-m depth were comparable to those observed in similar depths by others. During a 5-yr study in Georgia, 'Coastal' bermudagrass pastures were managed under a combination of nutrient sources and methods of defoliation (Franzluëbbers and Stuedemann 2005). Changes in soil C and N concentrations in the top 0.15-m depth ranged from 12.8 to 17.2 g C kg⁻¹ and 0.87 to 1.18 g N kg⁻¹, comparable with 14.3 g C kg⁻¹ and 1.1 g N kg⁻¹ for the 0- to 0.1-m depth for GN systems in the current study. During 12 yr, the same authors found soil C accumulation (0–0.15 m) under Coastal bermudagrass ranged from 0.08 to 1.34 Mg C ha⁻¹ yr⁻¹ (Franzluëbbers and Stuedemann 2009), a range that includes the values for both treatments in the current study. A range of near zero to 3.5 Mg C ha⁻¹ yr⁻¹ was reported for the 0- to 0.10-m depth for mixed-species temperate grasslands in northern England (De Deyn et al. 2011), and the 9-yr average was 0.83 Mg C ha⁻¹ yr⁻¹ for the 0- to 0.05-m stratum of 2- and 5-species forage mixtures in Pennsylvania, USA (Skinner and Dell 2016). The change in N stock in the current study was not affected by treatment and averaged 0.08 Mg N ha⁻¹ yr⁻¹, a value in the lower part of the annual range of zero to 0.35 Mg N ha⁻¹ yr⁻¹ reported for temperate forage mixtures (De Deyn et al. 2011).

An important question is what accounted for greater increases in soil C and N in surface soil of the GN vs the LG system. We propose that greater increases were likely due in part to greater aboveground herbage accumulation and residual biomass for GN than LG, and a trend toward a forage system effect on belowground RRM (3690 and 3020 kg OM ha⁻¹ for GN and LG, respectively) ($P = 0.123$). Contributions from both above- and belowground plant biomass is supported in the literature, and they are the main sources of input of organic material through decomposition and cycling processes. In grasslands, belowground biomass is often considered the dominant soil C supply because of the proportion of C from photosynthesis allocated belowground (Rasse et al.

2005; Sokol et al. 2019b). Living root inputs are thought to be much more efficient than aboveground litter inputs in forming both mineral-associated and particulate soil organic C (Sokol et al. 2019a). Recent research suggests that dissolved OM in leachate from aboveground litter enters the soil profile during precipitation events, providing a significant source of C, especially in high-leaching environments (Sokol et al. 2019b) like the sandy soils at the research site in the current experiment.

Despite differences in surface soil C and N accumulation, no treatment effects were observed in soil depths below 0.10 m. Similarly, Franzluebbers and Stuedemann (2005) observed no treatment responses in soil C at depths > 0.15 m, and in later work the same authors indicated that while surface accumulation of SOC and total soil N occurred, increased variability and loss of SOC with depth reduced the significance of surface effects (Franzluebbers and Stuedemann 2009). One interesting feature of the depth response in the current study was the proportional contribution to C stock of the top 0.10 m vs the 0.10- to 0.20 soil depth. When considering total C stock change in the combined 0- to 0.2-m soil depth over the 6 yr, 31% was attributable to change in the 0.1- to 0.2-m stratum for the GN system, but 79% was attributable to this stratum for LG. Five-species mixtures in Pennsylvania including alfalfa (*Medicago sativa* L.) accumulated more soil C at depths below 0.1-m than a two-species mixture including orchardgrass (*Dactylis glomerata* L.) and white clover (*Trifolium repens* L.) (Skinner and Dell 2016). They suggested greater root mass below 0.10 m for the mixture including alfalfa (760 vs. 560 kg ha⁻¹) contributed to the observed C accumulation response. In a separate Pennsylvania study, proportion of root biomass in the 0- to 0.05-m stratum decreased and proportion of root biomass in deeper soil strata increased with increasing species richness from two, to three, and to eleven species (Skinner et al. 2006).

The important role of root mass and root distribution in soil C accumulation has been observed by others (Cong et al. 2014), and it has been suggested that placing roots deeper in the soil profile can increase C accumulation relative to shallow roots as C stored deeper is less susceptible to mineralization and loss (Liebig et al. 2005). In a 30-yr grazing study, Somenahally et al. (2020) evaluated grass-N and legume-based systems managed under low (~ 680 kg

animal live weight ha⁻¹) or high (~ 2040 kg live weight ha⁻¹) grazing pressure. Microbial biomass was a major driver of the belowground C pool contributing between 35 and 55%. We did not quantify root-rhizome distribution among soil strata or microbial biomass in the current experiment, so it is not known if greater root mass or proportion at depth for LG than GN can explain the response observed.

For the current study, there was a general trend of C and N stocks increasing in the 0- to 0.4-m depth. Increases were mainly observed in the 0- to 0.2-m depth, while the 0.2- to 0.4-m depth had decreasing C stock during the 6 yr. Similarly, under a two-species mixture in Pennsylvania, the soil C loss in the strata below 0.05 m was almost half of the total C accumulated in the top 0.05-m depth over a 9-yr period (Skinner and Dell 2016).

There was no defoliation method effect on change in C stock in the 0.2- to 0.4-m stratum over the 6 yr (− 3.77 Mg C ha⁻¹ for the haying treatment and + 0.61 Mg C ha⁻¹ under grazing; $P = 0.155$). Previous research showed greater soil C accumulation for grazing than haying management, detectable after 12 yr but not after 5 yr (Franzluebbers and Stuedemann 2009). The pattern of response seen after 6 yr in the current study may be indicative of a long-term outcome similar to that observed by Franzluebbers and Stuedemann (2009).

Conclusions

The reported advantage in soil C accumulation of legume or legume-grass systems over grass-only systems (Wright et al. 2004; Jensen et al. 2012; Cong et al. 2014) is not guaranteed. Relative biomass productivity of systems, N fertilizer inputs to the grass system, and defoliation management will likely have a direct impact on C accumulation over time. High field variability in soil responses in the current study likely contributed to some differences not being detected, and maximizing replication is recommended in future experiments of this type. Future research should address (1) main drivers of SOC accumulation in multi-species forage systems including sub-tropical and tropical legumes, (2) differences in soil C and N accumulation as a function of soil depth for legume- and grass-based systems, and (3) the role played by N fertilization rates of grass monocultures on relative

comparisons of soil C and N accumulation of grass versus legume-grass systems. This information will allow a greater understanding of the processes involved and provide more accurate estimates of C budgets in forage-livestock systems.

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